

WILD RAMIFICATION AND REFINED SWAN CONDUCTORS AT 23 IN TWO EXPLICIT DEGREE-23 M_{23} -COVERS

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ABSTRACT. We study two explicit degree-23 covers with geometric monodromy group M_{23} : the rational three-point cover of Huang–Jackson–Lee–Poonen–Pries–Zhang and a polynomial cover of Elkies defined over a quartic field. After completing the two target function fields at explicitly defined Gauss valuations above 23, the induced degree-23 extensions have purely inseparable residue-field degree 23 and different of valuation 1, where $v(23) = 1$. We prove that the decomposition group in each Galois closure is the nontrivial semidirect product $C_{23} \rtimes C_{11}$, denoted 23:11, not C_{23} . Its quotient of order 11 gives a totally and tamely ramified extension; over that extension, the remaining cyclic extension of degree 23 is fierce in the standard sense of Wewers. Its Swan conductor is 23/22. We compute its refined Swan conductor from Wewers’s formula, as well as explicit degree-11 Kummer equations for the tame extensions. Finally, the action of the M_{23} normalizer on the 22 nontrivial characters of C_{23} has two orbits of size 11, whereas the corresponding S_{23} normalizer has one orbit of size 22. Exact PARI/GP and GAP computations certify the polynomial, valuation, norm, and finite-group calculations used in the proofs.

1. INTRODUCTION

The Mathieu group M_{23} is the 4-transitive sporadic simple subgroup of S_{23} of order

$$|M_{23}| = 2^7 \cdot 3^2 \cdot 5 \cdot 7 \cdot 11 \cdot 23.$$

Elkies classified the complex degree-23 polynomial covers with geometric monodromy M_{23} and gave explicit models over quartic number fields [2]. More recently, Huang, Jackson, Lee, Poonen, Pries, and Zhang constructed a regular M_{23} -extension over $\mathbf{Q}(T)$, meaning that $\mathbf{Q}$ is algebraically closed in the extension, and thereby realized M_{23} over $\mathbf{Q}$ [3]. Their degree-23 quotient has branch-cycle conjugacy classes (2A, 23A, 23B), in the standard *Atlas* notation, at

$$T = 0, \quad T = \sqrt{-23}, \quad T = -\sqrt{-23}.$$

In characteristic 23, both covers have wild ramification. The group theory relevant to this ramification is

$$N_{M_{23}}(C_{23}) = 23:11, \quad N_{S_{23}}(C_{23}) = 23:22.$$

The quotient of the first normalizer by C_{23} is the subgroup of nonzero squares in $\mathbf{F}_{23}^\times$; the quotient of the second is all of $\mathbf{F}_{23}^\times$.

This paper concerns the local field extensions obtained by completing at specified Gauss valuations. It does not construct a stable model, and it does not describe an entire special fiber or its components. For each cover we prove the following concrete statements:

- (i) the residue extension is purely inseparable of degree 23, and the different has normalized valuation 1;
- (ii) the decomposition group is 23:11;
- (iii) a factorization through a totally and tamely ramified degree-11 subextension, followed by a fierce cyclic extension of degree 23 with Swan conductor 23/22;

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- (iv) explicit degree-11 Kummer equations and explicit refined Swan conductors;
- (v) the orbit decomposition of the nontrivial C_{23} -characters under the M_{23} and S_{23} normalizers.

The comparison is useful because the covers are not two coordinate models of the same cover: their coefficient fields and branch-cycle descriptions are different. Nevertheless, they have the same decomposition group and the same Swan conductor at the valuations studied here.

The general framework is the stable reduction of three-point covers [8, 9] and Kato's refined Swan conductor for imperfect residue fields [4]. We use the explicit order- p formulas of Wewers [10, Proposition 5.3 and Corollary 5.4].

The division between prior work and the contribution here is as follows. The equations and geometric-monodromy results for the two covers are due to [2, 3]; the theory of fierce ramification and refined Swan conductors used below is due to Kato and Wewers; and the normalizer data for M_{23} are classical. The new content is the completed local calculation at the specified valuations: the different, the decomposition group, the tame degree-11 fields and their Kummer equations, and the resulting refined Swan conductors for both explicit covers. No general theorem about all M_{23} -covers is asserted.

Local terminology and conventions. All valuations below are normalized by $v(23) = 1$. If k is a complete discretely valued field and u is an indeterminate, the u -Gauss valuation on $k(u)$ is defined on polynomials by

$$v\left(\sum_i a_i u^i\right) = \min_i v(a_i)$$

and is extended to rational functions by taking differences of valuations. Its residue field is $\bar{k}(u)$.

For a finite separable extension L/K , the *different* $\mathfrak{D}_{L/K}$ is the usual different ideal of the extension of valuation rings. We call L/K *fierce*, following Wewers [10, Section 1.2], if $\bar{L}/\bar{K}$ is purely inseparable and

$$[L : K] = [\bar{L} : \bar{K}].$$

In particular, a fierce extension does not enlarge the value group. This standard term is needed because the usual ramification formulas for perfect residue fields do not apply directly.

We will also use initial forms. For a valued field K and $\gamma \in v(K^\times)$, set

$$\mathfrak{p}_K^\gamma = \{a : v(a) \geq \gamma\}, \quad \mathfrak{p}_K^{>\gamma} = \{a : v(a) > \gamma\}.$$

If $a \neq 0$ and $v(a) = \gamma$, then $\text{in}_v(a)$ denotes the class of a in $\mathfrak{p}_K^\gamma / \mathfrak{p}_K^{>\gamma}$, the degree- γ part of the associated graded ring. Thus an “initial-form equality” below is an ordinary equality in this associated graded ring. We use the same symbol for a uniformizer and for its initial form when no confusion can result.

Main result. Let f_H (with H for Huang et al.) denote the degree-23 cover from [3], in the exact model distributed with that work, and let f_E (with E for Elkies) denote the polynomial cover used in [2]. Let K_H and K_E be the completions at the Gauss valuations defined in Section 2. Let L_c/K_c be the induced degree-23 extension and M_c/K_c its Galois closure, for $c \in \{H, E\}$.

Theorem 1.1 (Ramification at 23). *For $c \in \{H, E\}$, the following statements hold.*

- (a) *The residue extension $\bar{L}_c/\bar{K}_c$ is purely inseparable of degree 23, the value group is unchanged, and*

$$v(\mathfrak{D}_{L_c/K_c}) = 1.$$

- (b) *The Galois group of the completed Galois closure is*

$$\text{Gal}(M_c/K_c) \cong 23:11.$$

If $P_c \cong C_{23}$ is its normal subgroup of order 23 and $K'_c = M_c^{P_c}$, then K'_c/K_c is totally and tamely ramified of degree 11, while M_c/K'_c is fierce and cyclic of degree 23.

- (c) The unique lower ramification break of M_c/K'_c is $1/22$. In Wewers's upper numbering, its Swan conductor is

$$\delta = \frac{23}{22}.$$

Fix a generator σ of P_c , choose $x \in O_{M_c}^\times$ whose residue generates $\bar{M}_c/\bar{K}'_c$, and set

$$b_c = N_{M_c/K'_c} \left(\frac{\sigma(x)}{x} - 1 \right), \quad y_c = N_{M_c/K'_c}(x).$$

If $\chi(\sigma) = 1/23$, then Kato's refined Swan conductor is

$$\text{rsw}(\chi) = b_c^{-1} \otimes \text{dlog}(\bar{y}_c).$$

In the two cases, the logarithmic differentials are

$$\text{dlog}(\bar{y}_H) = \text{dlog} \left(-\frac{E(z)}{(z-1)^5} \right), \quad \text{dlog}(\bar{y}_E) = \frac{dt}{t},$$

where E is given in Equation (2). Each differential is fixed by the Cartier operator. With the initial forms ξ_H and ξ_E defined in Equations (17) and (22), the coefficients in the refined Swan conductors have initial forms

$$\text{in}_v(b_H) = 8s^2 \xi_H \frac{(z-2)^8(z-1)}{E(z)}, \quad \text{in}_v(b_E) = \frac{14\pi^4 \xi_E}{t}.$$

- (d) Conjugation by the complement C_{11} acts on the nontrivial C_{23} -characters through the nonzero squares in $\mathbf{F}_{23}^\times$. It therefore has two orbits of size 11, indexed by squares and nonsquares. The normalizer of C_{23} in S_{23} has one orbit of size 22.

The proof occupies Sections 3 to 7. The short group-theoretic step is the exclusion of a cyclic Galois closure. If L_c/K_c were Galois with group C_{23} , all 22 nonidentity automorphisms would have the same lower ramification value. The different-product formula and $v(\mathfrak{D}_{L_c/K_c}) = 1$ would force that value to be $1/22$. But the value groups of K_H and K_E are $(1/2)\mathbf{Z}$ and $(1/4)\mathbf{Z}$, respectively, and neither contains $1/22$. Hence the decomposition group must contain the complement of order 11.

2. THE TWO COVERS AND THE GAUSS VALUATIONS

2.1. The rational three-point cover. The construction of [3] gives an exact degree-23 polynomial $F(V, T)$ over $\mathbf{Q}(T)$ whose Galois closure has group M_{23} . Its branch locus is $T(T^2 + 23) = 0$ with local monodromy $(2A, 23A, 23B)$. After reduction modulo 23, the three marked target points coincide in the published coordinate. Put

$$k_H = \mathbf{Q}_{23}(s), \quad s^2 = -23, \quad z = \frac{2T}{T+s}.$$

Then $T = 0, s, -s$ map to $z = 0, 1, \infty$. The unique extension of the 23-adic valuation to k_H has absolute ramification index $e_H = 2$. Let K_H be the completion of $k_H(z)$ for the z -Gauss valuation. Then

$$v(K_H^\times) = (1/2)\mathbf{Z}, \quad \bar{K}_H = \mathbf{F}_{23}(z).$$

The cover induces a degree-23 extension L_H/K_H .

After the integral change of source coordinate used in the exact verification script, the reduction of its monic defining polynomial is

$$\bar{F}_H(V, z) = V^{23} + \frac{E(z)}{(z-1)^5}, \tag{1}$$

where

$$\begin{aligned} E(z) = & 8z^9 + 7z^8 + 11z^7 + 16z^6 + 3z^5 + 19z^4 \\ & + 21z^3 + 21z^2 + 17z + 15 \in \mathbf{F}_{23}[z]. \end{aligned} \tag{2}$$

The polynomial E has a simple zero at $z = 1$, so the rational function in Equation (1) has pole order 4 there. It is therefore not a 23rd power in $\mathbf{F}_{23}(z)$. Thus Equation (1) is irreducible and defines a purely inseparable extension of degree 23.

The quadratic constant involution has a particularly small quotient on this residual equation.

Proposition 2.1 (Branch-exchange quartic). *Let ι be the nontrivial automorphism of $\mathbf{Q}_{23}(s)/\mathbf{Q}_{23}$, so that $\iota(s) = -s$ and $\iota(T) = T$. On the stable target coordinate it induces*

$$\tau(z) = \frac{z}{z-1}.$$

Put

$$u = \frac{z^2}{z-1}, \quad H(u) = 15(u^4 - u^3 + 8u^2 + 4u + 1) \in \mathbf{F}_{23}[u].$$

Then $\mathbf{F}_{23}(z)^{\langle \tau \rangle} = \mathbf{F}_{23}(u)$ and

$$-\frac{E(z)}{(z-1)^5} = H(u). \quad (3)$$

The polynomial map $H : \mathbf{P}_u^1 \rightarrow \mathbf{P}^1$ is tamely branched over $\infty, 18, 1$, with respective fibre partitions

$$(4), \quad (3, 1), \quad (2, 1, 1).$$

Its geometric monodromy group is S_4 . More precisely, after the affine changes

$$X = 4(u - 4), \quad W = 4(H - 18),$$

it has the Shabat normal form

$$W = X^3(4 - 3X). \quad (4)$$

Proof. The coordinate formula for τ follows immediately from $z = 2T/(T + s)$, and direct substitution gives

$$\tau^2(z) = z, \quad qquad(z-1)^{10}E\left(\frac{z}{z-1}\right) = E(z).$$

This proves the invariance in Equation (3). Moreover, z satisfies $z^2 - uz + u = 0$, whose other root is $\tau(z)$; hence u generates the fixed field.

The remaining assertions follow from the exact identities

$$\begin{aligned} H'(u) &= 14(u-4)^2(u-10), \\ H(u) - 18 &= 15(u-4)^3(u-12), \\ H(u) - 1 &= 15(u-10)^2(u^2 + 19u + 12). \end{aligned}$$

The last quadratic is separable, and $\deg H = 4$, giving the displayed passport. Since the characteristic is 23, all three branch indices are tame. A transitive degree-4 group containing both a 4-cycle and a 3-cycle is S_4 . Finally, substitution of the displayed X and W gives Equation (4). $\square$

Remark 2.2. The marked triple critical point of this quartic is intrinsic, but its displayed coordinate $u = 4$ is not: Equation (4) moves it to $X = 0$. The affine-invariant ratio formed from the triple critical point $u = 4$, the double critical point $u = 10$, and the simple companion $u = 12$ in the fibre over 18 is

$$\frac{12-4}{10-4} = 9 \quad \text{in } \mathbf{F}_{23}.$$

This quotient calculation is supplementary to Theorem 1.1; no rationality or stable-model assertion is deduced from the numerical coordinate values.

2.2. The Elkies polynomial cover. Elkies's polynomial is defined over

$$F_E = \mathbf{Q}[g]/(g^4 + g^3 + 9g^2 - 10g + 8).$$

After normalization, its finite branch values are $t = 0, 1$. In the degree-23 permutation action, their cycle shapes are $1^3 2^2 4^4$ and $1^7 2^8$, while the inertia generator over infinity is a 23-cycle [2]. There is a unique prime $\mathfrak{p}$ of F_E above 23; it has ramification index 4, residue degree 1, and uniformizer

$$\pi = g + 6.$$

Let $k_E = (F_E)_{\mathfrak{p}}$ be the completion, and let K_E be the completion of $k_E(t)$ for the t -Gauss valuation. Then

$$v(K_E^\times) = (1/4)\mathbf{Z}, \quad \bar{K}_E = \mathbf{F}_{23}(t).$$

The cover induces a degree-23 extension L_E/K_E . Writing $B(x)$ for the normalized polynomial map and using the source substitution $x = \pi^{-1}y$, reduction of the equation $B(x) = t$ gives

$$18y^{23} - t = 0. \quad (5)$$

Since t has pole order 1, it is not a 23rd power in $\mathbf{F}_{23}(t)$. This again gives a purely inseparable residue extension of degree 23.

Remark 2.3. The two covers have different coefficient fields and different branch-cycle descriptions, so they are not two coordinate presentations of the same cover. Their common Swan conductor below is therefore not a consequence of a change of coordinates between the two equations.

3. THE DIFFERENT AT THE GAUSS VALUATION

We continue to use $v(23) = 1$. When we give an integer exponent relative to s or π , we say so explicitly.

Lemma 3.1 (Monogenic normalization). *Let R be a henselian discrete valuation ring with fraction field K , and let $A = R[x]/(G)$ be finite and generically separable of prime degree p . Suppose that $A/\mathfrak{m}_R A$ is a field and is purely inseparable of degree p over the residue field of R . Then A is a discrete valuation ring and hence is the integral closure of R in its fraction field. Its different is generated by $G'(x)$.*

Proof. The reduction of G is irreducible because the special fiber is a field; hence G is irreducible and A is a domain. The same hypothesis shows that A is local with maximal ideal $\mathfrak{m}_R A$. This is a one-dimensional local domain whose maximal ideal is generated by a uniformizer of R . It is therefore a regular local ring of dimension one, hence a discrete valuation ring. Since a discrete valuation ring is integrally closed, A is the integral closure of R in its fraction field. The derivative formula for the different of a monogenic separable extension is standard; see, for example, [5, Chapter III]. $\square$

3.1. The rational cover. Let $G_H(V, z)$ be the integral monic equation whose reduction is Equation (1). Exact differentiation shows that $G'_H = 23\tilde{\Delta}_H$ with $\tilde{\Delta}_H$ integral. Let Δ_H be its reduction. This polynomial is nonzero of degree 22 in V . Since the residue of V has degree 23 over $\mathbf{F}_{23}(z)$, Δ_H does not vanish at that residue class. Its norm to $\mathbf{F}_{23}(z)$ is

$$N(\Delta_H) = \frac{z^{30}(z-2)^{102}(z^2+13z+10)^{22}}{(z-1)^{88}}. \quad (6)$$

The quadratic factor is irreducible over $\mathbf{F}_{23}$: its discriminant is 14, which is a nonsquare modulo 23.

By Lemma 3.1, the integral source algebra is the full integral closure. At its generator, G'_H is 23 times a unit. Hence

$$v(\mathfrak{D}_{L_H/K_H}) = 1.$$

Since $v(s) = 1/2$, the exponent relative to the uniformizer s is 2. Taking the norm to the base multiplies this exponent by the residue degree 23, so the discriminant exponent relative to s is 46.

3.2. **The Elkies cover.** For $x = \pi^{-1}y$, direct exact reduction gives

$$\pi^{-4} \frac{dB(\pi^{-1}y)}{dy} \equiv -y^{14}(y-9)^8 \pmod{\pi}. \quad (7)$$

Under $t = 18y^{23}$, the two zeros $y = 0, 9$ map to $t = 0, 1$. The exact resultant is

$$N(-y^{14}(y-9)^8) = -t^{14}(t-1)^8. \quad (8)$$

The multiplicities have the expected interpretation in the degree-23 action. For cycle shape $1^3 2^2 4^4$, the sum of $e_i - 1$ is $2(2-1) + 4(4-1) = 14$; for $1^7 2^8$, it is 8. Again Lemma 3.1 applies. Equation (7) says that the derivative is π^4 times a unit at the residue class of y . Thus the different exponent relative to π is 4, or equivalently $v(\mathfrak{D}_{L_E/K_E}) = 1$. The discriminant exponent relative to π is $23 \cdot 4 = 92$.

The two normalizations are summarized in the following table. The last two columns use the displayed uniformizer, not the convention $v(23) = 1$.

TABLE 1. Different and discriminant exponents at the two Gauss valuations.

cover	uniformizer	$e(k/\mathbf{Q}_{23})$	$[\bar{L} : \bar{K}]$	$\text{ord}_{\varpi}(\mathfrak{D})$	$\text{ord}_{\varpi}(\text{disc})$
rational three-point	s	2	23	2	46
Elkies polynomial	π	4	23	4	92

4. THE DECOMPOSITION GROUP

Fix either cover and suppress the subscript c . Choose a valuation of the global Galois closure above the Gauss valuation, and let M be the corresponding completion. Then $D = \text{Gal}(M/K)$ is the decomposition group. The local degree-23 polynomial is irreducible, so D acts transitively on its 23 roots. Hence 23 divides $|D|$. The 23-part of $|M_{23}|$ has order 23, so a Sylow 23-subgroup P of D is cyclic and acts as a 23-cycle. It is the wild ramification subgroup; this subgroup is canonical and therefore normal in D . Consequently

$$P \triangleleft D \subseteq N_{M_{23}}(P) = 23:11.$$

Since 11 is prime, there are only two possibilities:

$$D = P \quad \text{or} \quad D = 23:11.$$

Proposition 4.1 (Value-group obstruction). *For both covers, the possibility $D = P$ is impossible.*

Proof. Assume $D = P$. Then the degree-23 extension L/K is itself Galois. It is fierce: its residue extension is purely inseparable of degree 23, its field degree is 23, and its value group is unchanged.

Let x be the integral source generator used above. Its residue generates $\bar{L}/\bar{K}$, so $O_L = O_K[x]$ by Lemma 3.1. For $\sigma \neq 1$ in P , put $i(\sigma) = v(\sigma(x) - x)$. For a monogenic fierce extension, the sets defined by lower bounds on $i(\sigma)$ are the lower ramification subgroups [10, Section 3.1]. Since a group of prime order has no proper nontrivial subgroup, all 22 nonidentity values $i(\sigma)$ are equal. The factorization of the derivative at x gives

$$1 = v(\mathfrak{D}_{L/K}) = \sum_{\sigma \neq 1} v(\sigma(x) - x) = 22i(\sigma).$$

Therefore $i(\sigma) = 1/22$.

For the rational cover the value group is $(1/2)\mathbf{Z}$, while for the Elkies cover it is $(1/4)\mathbf{Z}$. Neither contains $1/22$. This contradicts $\sigma(x) - x \in L$, and excludes $D = P$. $\square$

Corollary 4.2. *For both covers, $D = 23:11$. If $K' = M^P$, then K'/K is totally tamely ramified of degree 11, and M/K' is fierce cyclic of degree 23.*

Proof. Only 23:11 remains. Label the roots r_j , $j \in \mathbf{F}_{23}$, so that P acts by translations. A point stabilizer C_{11} acts on ordered differences $r_{j+a} - r_j$ through multiplication of a by a nonzero square. It therefore has two orbits, indexed by the squares and nonsquares in $\mathbf{F}_{23}^\times$. Reversing an ordered pair replaces a by $-a$ and does not change the valuation. Since -1 is a nonsquare modulo 23, reversal exchanges the two orbits. Thus every root difference has a common valuation i . At any root the different-product formula gives

$$1 = \sum_{a \neq 0} v(r_a - r_0) = 22i,$$

so $i = 1/22$.

The value group of M contains the value group of K and the value $1/22$. The subgroup generated by these values is $(1/22)\mathbf{Z}$ in the rational case and $(1/44)\mathbf{Z}$ in the Elkies case. In either case its index over $v(K^\times)$ is 11, so the ramification index $e(M/K)$ is at least 11. On the other hand, M contains L , whose residue field has degree 23 over $\bar{K}$. The fundamental inequality $[M : K] \geq e(M/K)[\bar{M} : \bar{K}]$ and $[M : K] = 253$ now give

$$e(M/K) = 11, \quad [\bar{M} : \bar{K}] = 23.$$

In particular, there is no defect in M/K : equality holds in the fundamental inequality. Since $[K' : K] = 11$ and $[M : K'] = 23$, multiplicativity of ramification indices shows that $e(K'/K) = 11$ and $e(M/K') = 1$. Thus K'/K is totally ramified and tame, with trivial residue extension. The extension M/K' has unchanged value group and purely inseparable residue-field degree 23; by definition it is fierce. $\square$

Remark 4.3. The extension K'/K changes the value group but not the residue field. It is not an unramified extension of degree 11. This distinction matters in the Swan-conductor calculation.

5. EQUATIONS FOR THE TAME EXTENSIONS OF DEGREE 11

We first give the common construction. Label the roots r_j , $j \in \mathbf{F}_{23}$, so that a generator of P acts by $r_j \mapsto r_{j+1}$. Let

$$Q = (\mathbf{F}_{23}^\times)^2 = \{1, 2, 3, 4, 6, 8, 9, 12, 13, 16, 18\}, \quad (9)$$

and put

$$C_a = \prod_{j \in \mathbf{F}_{23}} (r_{j+a} - r_j), \quad A = \prod_{q \in Q} C_q, \quad (10)$$

for $a \in \mathbf{F}_{23}^\times$. Translation fixes every C_a , so $C_a \in K' = M^P$. The complement C_{11} permutes the eleven elements C_q with $q \in Q$. Hence

$$\mathcal{R}_C(U) = \prod_{q \in Q} (U - C_q) \in K[U] \quad (11)$$

is a degree-11 resolvent: its coefficients are invariant under the full decomposition group. Its constant term is $-A$.

Since the degree is odd, $C_{-a} = -C_a$. Moreover, -1 is a nonsquare modulo 23, so the nonzero indices are the disjoint union $Q \sqcup (-Q)$. For a monic polynomial f ,

$$\prod_{a \neq 0} C_a = \prod_j f'(r_j) = \text{Res}(f, f') = -\text{Disc}(f),$$

whereas the left side is $-A^2$. Thus

$$A^2 = \text{Disc}(f). \quad (12)$$

Replacing the chosen generator of P by a nonsquare power exchanges Q and $-Q$ and replaces A by $-A$. Thus, in each explicit formula below, we choose the generator of P so that A equals the displayed square root of the discriminant.

Lemma 5.1 (Initial forms of the root differences). *For $a \in \mathbf{F}_{23}^\times$ and $j \in \mathbf{F}_{23}$, one has*

$$\mathrm{in}_v(r_{j+a} - r_j) = a \, \mathrm{in}_v(r_{j+1} - r_j).$$

Consequently C_a/C_1 is a unit with residue a , and A/C_1^{11} is a principal unit, meaning a unit with residue 1.

Proof. Put $d_j = r_{j+1} - r_j$. By Corollary 4.2, all d_j have value $i = 1/22$. The group P acts trivially on the residue field of M : that residue extension is purely inseparable over the residue field of K' , and hence has no nontrivial automorphisms. It follows that the action of P on the one-dimensional $\bar{M}$ -vector space

$$\mathfrak{p}_M^i / \mathfrak{p}_M^{>i}$$

is linear. Every eigenvalue of this action is a 23rd root of unity in the characteristic-23 field $\bar{M}$, and therefore is 1. In particular, $\mathrm{in}_v(d_j)$ is independent of j .

Represent a by an integer in $\{1, \dots, 22\}$. The telescoping identity

$$r_{j+a} - r_j = d_j + d_{j+1} + \dots + d_{j+a-1}$$

then gives the asserted initial-form equality; the scalar a is nonzero in $\mathbf{F}_{23}$, so no cancellation occurs in the lowest nonzero graded piece. Multiplication over j gives

$$\overline{C_a/C_1} = a^{23} = a.$$

Finally, Q is the group of roots of $X^{11} - 1$ in $\mathbf{F}_{23}$, so $\prod_{q \in Q} q = 1$. Therefore $\overline{A/C_1^{11}} = 1$, as required. $\square$

We next replace this resolvent by a Kummer equation. By Lemma 5.1, A/C_1^{11} is a principal unit. Because 11 is invertible in the valuation ring, Hensel's lemma gives a unique principal unit u satisfying $u^{11} = A/C_1^{11}$. Consequently

$$\beta = uC_1, \quad \beta^{11} = A, \quad \mathrm{in}_v(\beta) = \mathrm{in}_v(C_1). \quad (13)$$

The residue field contains all eleventh roots of unity because $11 \mid |\mathbf{F}_{23}^\times|$, and Hensel's lemma lifts them to K . Moreover, $v(\beta) = 23/22 \notin v(K^\times)$. The class of this value has exact order 11 modulo $v(K^\times)$: indeed, $11v(\beta) = 23/2$ belongs to both $(1/2)\mathbf{Z}$ and $(1/4)\mathbf{Z}$, while $v(\beta)$ belongs to neither. Hence $[K(\beta) : K] \geq 11$. Since $\beta \in K'$ and $[K' : K] = 11$, the element β generates K'/K , and

$$X^{11} - A$$

is its irreducible Kummer equation. Only the initial form of β is needed later: the refined Swan conductor is tensored with a residue-field differential, so multiplying its coefficient by a principal unit does not change the resulting class.

5.1. The rational three-point cover. Let $G_{84}(T)$ be the primitive degree-84 polynomial appearing in the exact identity

$$\mathrm{Disc}_V(F) = \left(\frac{2^{105} 3^{36} 23^{37} T^4 G_{84}(T)}{(T^2 + 23)^{44}} \right)^2. \quad (14)$$

Choose the generator of P so that A is the square root displayed inside the parentheses in Equation (14). Set

$$h_H = \frac{2^9 3^3 23^5}{(T^2 + 23)^4}, \quad \Lambda_H = \frac{2^6 3^3 T^4 G_{84}(T)}{23^{18}}. \quad (15)$$

With this choice of generator, the displayed square root is the element A of Equation (10), and it is $h_H^{11} \Lambda_H$. Thus, after replacing β by $\theta_H = \beta/h_H$, the tame extension is generated by a root of

$$R_H(X) = X^{11} - \Lambda_H. \quad (16)$$

Here v_s denotes the integer-valued valuation with $v_s(s) = 1$. Under $T = sz/(2 - z)$ one has $v_s(\Lambda_H) = 1$, so this polynomial is Eisenstein. Writing $w = z/(2 - z)$, reduction after division by s gives

$$\overline{\Lambda_H/s} = w^{15}(w^2 + 6)^{11} = \frac{z^{15}(z^2 + 13z + 10)^{11}}{(z - 2)^{37}}.$$

We therefore set $\xi_H = \text{in}_v(\theta_H)$ by

$$\xi_H^{11} = s \frac{z^{15}(z^2 + 13z + 10)^{11}}{(z - 2)^{37}}. \quad (17)$$

Write the constant coefficient of $F(V, T)$ as $b_{23}(T)/(T^2 + 23)^5$. The norm of a root is

$$Y_H = \prod_j r_j = -\frac{b_{23}(T)}{(T^2 + 23)^5}, \quad \bar{Y}_H = -\frac{E(z)}{(z - 1)^5}.$$

Finally, $\overline{h_H/s^2} = 15(z - 2)^8/(z - 1)^4$. Combining [Equations \(13\)](#) and [\(17\)](#) determines the initial form of b_H . Indeed, $b_H = C_1/Y_H$, while β/C_1 is a principal unit, so

$$\text{in}_v(b_H) = \text{in}_v\left(\frac{h_H\theta_H}{Y_H}\right) = 8s^2\xi_H \frac{(z - 2)^8(z - 1)}{E(z)}. \quad (18)$$

5.2. The Elkies polynomial cover. Write Elkies's normalized polynomial map as $B(x) = P(x)/\tau$, where $P \in F_E[x]$ has degree 23 and $\tau \in F_E^\times$, as in the exact input file. Put $p_{23} = \text{lc}_x(P)$, $\pi = g + 6$, and

$$s_E = \frac{2g^3 + 4g^2 + 16g - 7}{3}, \quad s_E^2 = -23. \quad (19)$$

Choose the generator of P_E so that the corresponding element A_E is the square root displayed below. For the monic local polynomial

$$f_E(y, t) = \frac{P(\pi^{-1}y) - \tau t}{p_{23}\pi^{-23}},$$

the exact discriminant square root is

$$A_E = s_E \left(\frac{23\tau\pi^{23}}{p_{23}} \right)^{11} t^7(t - 1)^4. \quad (20)$$

Indeed, the derivative has critical multiplicities 14 over 0 and 8 over 1, while its leading coefficient is $23p_{23}\pi^{-23}$; the resultant formula gives [Equation \(20\)](#) after using $s_E^2 = -23$.

Set

$$h_E = \frac{23\tau\pi^{23}}{p_{23}}, \quad \Lambda_E = s_E t^7(t - 1)^4.$$

Then $A_E = h_E^{11}\Lambda_E$ and the exact tame extension is generated by a root of

$$R_E(X) = X^{11} - \Lambda_E. \quad (21)$$

Here v_π is the integer-valued valuation with $v_\pi(\pi) = 1$. Since $v_\pi(\Lambda_E) = 2$ is coprime to 11, the equation gives a totally ramified extension of degree 11. Reduction after division by π^2 gives

$$\overline{\Lambda_E/\pi^2} = -3t^7(t - 1)^4.$$

Set $\theta_E = \beta/h_E$ and $\xi_E = \text{in}_v(\theta_E)$. Then

$$\xi_E^{11} = -3\pi^2 t^7(t - 1)^4. \quad (22)$$

The exact root norm and its reduction are

$$Y_E = \frac{\tau t - P(0)}{p_{23}\pi^{-23}}, \quad \bar{Y}_E = \frac{t}{18}.$$

Since $\overline{h_E/\pi^4} = 11$, we obtain, in the same way as in the rational case (using $b_E = C_1/Y_E$ up to the principal-unit replacement of C_1 by β),

$$\mathrm{in}_v(b_E) = \mathrm{in}_v\left(\frac{h_E\theta_E}{Y_E}\right) = \frac{14\pi^4\xi_E}{t}. \quad (23)$$

These formulas use the chosen generator σ of P . Replacing σ by a nonzero power changes the representative refined Swan conductor as described in [Section 7](#); it does not change the two normalizer orbits.

6. THE SWAN CONDUCTOR AND ITS REFINEMENT

We now apply Wewers's ramification theory for fierce cyclic extensions [[10](#), Sections 3 and 5]. The residue fields here are rational function fields k over $\mathbf{F}_{23}$ and satisfy $[k : k^{23}] = 23$, as required in that theory. For a fierce extension of degree p whose nonidentity automorphisms have common lower value i , the Herbrand function has slope p on the interval from 0 to i . Thus the upper break, which is also the Swan conductor of a faithful character, is pi . In our case

$$i = \frac{1}{22}, \quad \delta = 23i = \frac{23}{22}.$$

The value $23/22$ equals $p/(p-1)$ for $p = 23$, the largest value allowed by [[10](#), Corollary 5.4]. The same corollary says that the differential appearing in the refined Swan conductor is logarithmic: it has the form $d\bar{y}/\bar{y}$ and is fixed by the Cartier operator C .

We recall the exact coefficient of this differential. Fix a generator σ of P , let $\chi(\sigma) = 1/23$, and choose $x \in \mathcal{O}_M^\times$ such that $\bar{x}$ generates $\bar{M}/\bar{K}'$. Put

$$a_\sigma = \frac{\sigma(x)}{x} - 1, \quad b = N_{M/K'}(a_\sigma), \quad y = N_{M/K'}(x).$$

Then $v(b) = 23/22$, and formula (5) in the proof of [[10](#), Proposition 5.3] gives the equality

$$\mathrm{rsw}(\chi) = b^{-1} \otimes \frac{d\bar{y}}{\bar{y}} = b^{-1} \otimes \mathrm{dlog}(\bar{x}^{23}). \quad (24)$$

Wewers derives this formula after an auxiliary constant extension containing the required roots of unity. His invariance result [[10](#), Lemma 3.8] identifies the resulting refined Swan conductor with the one over the original field. More explicitly,

$$\mathrm{rsw}(\chi) \in \mathfrak{p}_{K'}^{-23/22} \otimes_{\mathcal{O}_{K'}} \Omega_{\bar{K}'}^1.$$

Because the maximal ideal of $\mathcal{O}_{K'}$ acts trivially on $\Omega_{\bar{K}'}^1$, the coefficient b^{-1} in this tensor is determined by $\mathrm{in}_v(b)$. Equations (18) and (23) compute that initial form for the two covers.

For the rational cover, [Equation \(1\)](#) gives

$$\bar{y} = \bar{x}^{23} = -\frac{E(z)}{(z-1)^5},$$

while [Equation \(5\)](#) gives $\bar{y} = \bar{x}^{23} = t/18$ for the Elkies cover. Therefore the differentials in [Equation \(24\)](#) are

$$\eta_H = \mathrm{dlog}\left(-\frac{E(z)}{(z-1)^5}\right), \quad \eta_E = \frac{dt}{t}.$$

They satisfy $C(\eta_H) = \eta_H$ and $C(\eta_E) = \eta_E$. Together with [Equations \(18\) and \(23\)](#), this proves [Theorem 1.1\(c\)](#).

7. NORMALIZER ORBITS ON THE NONTRIVIAL CHARACTERS OF C_{23}

Fix a generator σ of $P = C_{23}$. Conjugation gives an injection

$$N_{M_{23}}(P)/P \hookrightarrow \text{Aut}(P) \cong \mathbf{F}_{23}^\times.$$

The image has order 11, so it is the unique index-2 subgroup $Q = (\mathbf{F}_{23}^\times)^2$ listed in Equation (9). For $a \in \mathbf{F}_{23}^\times$, let χ_a be the character defined by $\chi_a(\sigma) = a/23$. If $\rho = \text{rsw}(\chi_1)$, then $\text{rsw}(\chi_a) = a\rho$; this is the standard behavior of the refined Swan conductor under multiplication of a character by a [10, Section 4.2]. Consequently the normalizer has two orbits on the 22 nontrivial characters:

$$\{\chi_a : a \in Q\}, \quad \{\chi_a : a \in \mathbf{F}_{23}^\times \setminus Q\}.$$

The same square/nonsquare partition describes the two normalizer orbits on the nonidentity elements of P ; these are the intersections of P with the two M_{23} conjugacy classes 23A and 23B. Choosing a different generator can interchange the two labels, but not the unordered pair of orbits.

For S_{23} , the normalizer quotient is all of $\mathbf{F}_{23}^\times \cong C_{22}$, so it is transitive on the 22 nontrivial characters. This is the precise local distinction established here: the M_{23} normalizer preserves the square/nonsquare partition, whereas the S_{23} normalizer does not. We do not construct an Artin or Euler factor from this partition.

The group assertions were checked independently in GAP. The calculation returns

ambient group	normalizer order	centralizer order	orbits on $P \setminus \{1\}$
M_{23}	253	23	11 + 11
S_{23}	506	23	22

and identifies the normalizer exponents—the elements $a \in \mathbf{F}_{23}^\times$ for which conjugation sends σ to σ^a —with the set in Equation (9). These values also follow from the standard group data in the *Atlas* [1].

8. EXACT CERTIFICATION AND REPRODUCIBILITY

The computation is split into an algebraic certificate and a finite-group certificate so that the two parts can be checked independently.

8.1. Algebraic certificate. The PARI/GP script

`scripts/certify_p23_vertical_different.gp`

loads fixed copies of the two exact cover equations, verifies the primes above 23 and their ramification indices, performs the coordinate changes, checks the valuations of the derivative coefficients, and proves the norm identities Equations (6) and (8). It also checks irreducibility of the quadratic factor in Equation (6) and nonvanishing of the two logarithmic differentials in Section 6. Its output table has SHA-256 digest

`f8f13dbab17744a71d5309ba82e591c4383797a48dadebbfb5540d69198b1fb9.`

The second PARI/GP script is

`scripts/certify_p23_tame_resolvents.gp`

It computes both discriminants, selects the square root A defined by Equation (10), removes the displayed eleventh powers, and verifies the two Kummer equations. It also checks the root norms and the initial-form identities Equations (18) and (23). Its output table has SHA-256 digest

`276d6089292d54834572c1ffab8be43279cb229bca8cb83b9b8590c48da79649.`

The third PARI/GP script is

`scripts/certify_p23_semilinear_covariance.gp.`

It verifies the branch-exchange functional equations, the fixed-field coordinate and quartic factorization in Proposition 2.1, the critical passport, and the Shabat normal form. Its output table has SHA-256 digest

`77ff9f2e4c9960c3e430a4b116e7cc3f1fb419a38d89fd8671e0be35bba09fa7.`

8.2. Group certificate. The GAP script

`scripts/certify_p23_stable_group.g`

constructs M_{23} in its natural action, computes a Sylow 23-subgroup, its normalizer and centralizer in both M_{23} and S_{23} , and computes the orbits and conjugation exponents. The output table has SHA-256 digest

`a1d0fc6a994efc043bef515eb881907385063e112953cb867f921cc5e41de78a.`

The original certificates were run with PARI/GP 2.17.4 and GAP 4.16.0 [7, 6]. A second implementation of all four certificate streams is supplied in `verification/magma/`. Those files are self-contained, are deterministically generated from the same fixed cover inputs, and terminate before their final PASS marker if any exact check fails. All four were run with Magma V2.29-9 on the University of Sydney public calculator; their input hashes, seeds, timings, and PASS markers are recorded in `verification/magma_verification_summary.json`. This provides a cross-implementation check of the exact computational identities; the deduction of the ramification theorem from those identities remains the mathematical argument of the preceding sections.

The publication PDF, exact certificates, and fixed cover inputs are available in the public companion repository

<https://github.com/research-reflexivity/frontier-math-m23-ramified-extension>.

From the repository root, the original certificate set is reproduced by

`make verify.`

The independent Magma suite is reproduced by

`make verify-magma.`

9. SCOPE OF THE RESULT

The paper proves statements about two completed local field extensions at the Gauss valuations defined in Section 2. It does not construct a semistable or stable model of either cover. In particular, it does not identify the irreducible components, intersection points, or tails of a special fiber. Such a construction would be needed before applying a vanishing-cycles formula to the whole cover.

The refined Swan conductor in Equation (24) is intrinsic to the local character. The displayed formulas Equations (18) and (23), however, use the chosen target coordinates and the chosen generator of P . Comparing divisors for the two covers would first require a specified identification of their residue function fields; no such identification is made here.

Finally, Section 7 proves an orbit statement for local characters. It does not attach Euler factors or an L -function to those orbits, and it does not assert that the same local calculation holds for every M_{23} -cover.

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